

Jahangirnagar University



Assignment On

Software Quality Factors (McCall's Quality Model)

Course Title: Software Project Management and Quality Assurance
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SUBMITTED TO

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Part - A

1. Why are software quality require considered non-functional requirements?

Answer: Functional requirements define what a system must do. Quality requirements are non-functional because they define how well the system performs those functions (e.g., how fast, how securely or how reliably the payment is processed).

2. What is the difference between functional requirements and quality factors?

Answer: Functional Requirements: specify the exact behavior, inputs and outputs of a system.
Quality factors: specify the constraints and performance standards of the system.

3. List three categories of McCall's software Quality Factors?

(P.T.O)

- product operation: How well the software runs.
- product revision: How easily the software can be changed.
- product transition: How easily the software can be moved or integrated.

4. Define correctness with an example?

Answer: Correctness is the extent to which a program fulfill its specifications and meets the client's objectives.

Example: If an E-KYC (Electronic-know your customer) onboarding system is specify to perfectly extract text from an uploaded National Id card using OCR to achieving accurate extraction without data loss demonstrates correctness.

5. What is meant by reliability? Give one measurable reliability requirement?

Answer: Reliability refers to the software

ability to provide continuous service without failure over a specified period.

measurable example: the backend API must contain 99.9% uptime resulting in no more than 43 minutes of downtime per month.

6. Explain the role of Integrity in software quality?

Answer: Integrity dictates the system's security and its ability to prevent unauthorized access to data. In an enterprise system, high integrity ensures that sensitive customer information remains encrypted and safe from breaches.

7. What is the difference between maintainability and flexibility?

Answer: Maintainability: the effort required to locate and fix bug or error in the software

Flexibility: the effort to modify or add new features to the software to adapt to changing business needs.

8. Why is testability important during software development?

Answer: Testability ensures that developers can easily verify if a module works correctly or if a bug has been successfully resolved. Software designed with good logging and diagnostic tools saves significant time during debugging and QA cycle.

9. Define portability with a practical example?

Answer: Portability is the ease of which software can be transferred from one hardware to another or software to another.

Example: Successfully maintaining a Java Spring boot backend database from one premise local server to an AWS cloud environment without having to rewrite the core application logic.

10. What is the interoperability and why is it important in enterprise software?

Answer: Interoperability is the ability of a system to connect and exchange data smoothly with other distinct systems. It is vital in enterprise environments because modern applications rarely operate in isolation (e.g., a geolocation based attendance application must seamlessly push data into a separate centralized HRMS API).